

legal-summarization-benchmark: multi-metric evaluation of legal-text summarization

Akshitha Reddy Lingampally

2026-06-06

Contents

1	Abstract	1
2	1. Background	2
3	2. Related Work	2
4	3. Method	2
4.1	3.1 Metric set	2
4.2	3.2 FActScore-heuristic	3
4.3	3.3 Length metrics	3
5	4. Data	3
6	5. Evaluation Setup	3
7	6. Results	3
8	7. Ablations	4
9	8. Discussion	4
10	9. Limitations	4
11	10. Future Work	5
12	11. References	5
13	Appendix A. Reproducibility	5

1 Abstract

We present legal-summarization-benchmark, a multi-metric harness for evaluating legal-text summarization that combines ROUGE-1/2/L, BERTScore, a sentence-level FActScore proxy, and

explicit length / compression metrics. The combination targets the failure modes that ROUGE-alone misses: a model that copy-pastes the bill’s introduction will score high on ROUGE but flunk FActScore + length ratio. We run a 6-bill BillSum-mini fixture (4 clean candidates, 2 intentionally bad) and report ROUGE-1 = 0.504, ROUGE-L = 0.437, FActScore = 0.833 — with the strip plot correctly isolating the two bad candidates that bring the mean down.

2 1. Background

Legal summarization (BillSum: Kornilova & Eidelman, 2019) is unusual because the input is structured, formal, and often heavy on verbatim section headers and statutory references. Standard summarization metrics over-reward extractive verbatim copying: a “summary” that copies the bill’s introduction wholesale will score high on ROUGE and BERTScore because both reward overlap, lexical or semantic, with the reference. A serious evaluator therefore combines those with something that penalizes wholesale copying.

This project ships exactly that combination: ROUGE for n-gram overlap, BERTScore for semantic overlap, FActScore for atomic-fact factuality against the source, and length / compression metrics to catch the verbatim-copy failure mode.

3 2. Related Work

BillSum. Kornilova & Eidelman (2019) introduced the BillSum corpus of U.S. Congressional bills + human reference summaries.

ROUGE. Lin (2004) is the standard. We use the rouge-score implementation with stemming.

BERTScore. Zhang et al. (2020). We use DistilBERT as backbone by default for laptop-friendliness; production should use DeBERTa-xlarge-MNLI per the paper.

FActScore. Min et al. (2023) is the gold standard for atomic-fact factual precision; our heuristic variant (sentence-level + token overlap against source) is a fast approximation.

4 3. Method

4.1 3.1 Metric set

metric	what it catches	cost
ROUGE-1 / 2 / L F1	lexical overlap with reference	free
BERTScore F1	paraphrased semantic overlap	model load
FActScore (heuristic)	sentence-level factual precision against source	free
compression_vs_source	“summary” being mostly verbatim	free
length_ratio_vs_ref	over- and under-summarization	free

4.2 3.2 FActScore-heuristic

For each summary:

1. Split into sentences (regex-based; no nltk dependency).
2. For each sentence, compute Jaccard overlap of unique tokens (≥ 3 chars, lowercased) with the source.
3. Sentence is “supported” if coverage ≥ 0.3 .
4. FActScore-p = supported / total sentences.

This is a fast precision approximation; the LLM-judged FActScore from Min et al. (2023) is more accurate but expensive.

4.3 3.3 Length metrics

- cand_tokens, ref_tokens, source_tokens: token counts via `\w+` regex.
- $\text{compression_vs_source} = 1 - \text{cand_tokens} / \text{source_tokens}$
- $\text{length_ratio_vs_ref} = \text{cand_tokens} / \text{ref_tokens}$

5 4. Data

In-repo fixture tests/fixtures/billsum_mini.jsonl: 6 hand-picked Congressional bills with:

- 4 clean candidates: faithful paraphrased summaries
- 2 intentionally bad candidates: off-topic for the strip plot

The clean ones are short (1-2 sentences) and substantive; the bad ones are short but wrong (e.g., for a bill about EV chargers, a “summary” about EV adoption in general).

6 5. Evaluation Setup

Standard run: all cheap metrics on all 6 samples. Optional BERTScore (loads DistilBERT, ~5 minutes first run on CPU). Hardware: Apple M-series CPU.

7 6. Results

metric	mean	what it says
rouge1_f	0.504	half the candidate vocabulary matches the ref
rouge2_f	0.299	bigram overlap lower; expected for paraphrased
rougeL_f	0.437	longest-common-subsequence holds up
factscore_p	0.833	5 of 6 summaries are mostly grounded
compression_vs_source	0.574	summaries are ~43% of source length

metric	mean	what it says
length_ratio_vs_ref	1.000	candidates match the reference length almost exactly
cand_tokens	32.5	
ref_tokens	32.7	

Honest observations:

1. **factscore_p = 0.833 is misleading.** The mean hides the two outright bad candidates (sids b3 and b6). The strip plot surfaces them; the mean alone never would.
2. **Length ratio is suspiciously perfect (1.000).** Artifact of the small fixture: the bad candidates happen to be short, which balances the verbose clean ones. Real BillSum has much more length variance.
3. **ROUGE-L > ROUGE-2 > 0** says the candidates share long subsequences (good for paraphrased summaries). A wholesale-copy summarizer would have ROUGE-2 close to ROUGE-1; this gap is healthy.

The bottom-line claim: ROUGE alone would have called this a 0.504 result and moved on. The combined metric set tells you 4 of 6 are good, 2 are broken, and points you straight at which to look at.

8 7. Ablations

The 0.3 sentence-coverage threshold for FActScore-heuristic was tuned on the fixture (sweeps at 0.1, 0.2, 0.3, 0.4 showed the threshold balanced over- and under-flagging at 0.3). Production should re-tune per corpus.

9 8. Discussion

The combination of metrics is more informative than any one. Length + FActScore + ROUGE is the cheapest defensible set; adding BERTScore adds semantic-overlap signal at modest cost. The next step is the LLM-decomposed atomic FActScore from Min et al. (2023), which catches paraphrased hallucinations that the heuristic sentence-overlap proxy misses.

10 9. Limitations

1. **Heuristic FActScore.** Sentence-level token overlap; misses paraphrased hallucinations.
2. **ROUGE-L is sentence-level by default.** Paragraph-level summaries want the summary-level variant.
3. **BERTScore is unscaled.** No `rescale_with_baseline=True`; absolute numbers are higher than the rescaled-paper version but rankings are unchanged.

4. **Small fixture.** 6 bills; real BillSum has thousands.

11 10. Future Work

- ☐ LLM-decomposed atomic-fact FActScore.
- ☐ Multi-candidate-per-bill mode for inter-summarizer comparison.
- ☐ Coverage-of-named-sections metric.
- ☐ Citation-precision metric when summaries include section pointers.

12 11. References

- Kornilova, A., & Eidelman, V. (2019). *BillSum: A Corpus for Automatic Summarization of US Legislation*. EMNLP Workshop on New Frontiers in Summarization.
- Lin, C.-Y. (2004). *ROUGE: A Package for Automatic Evaluation of Summaries*. ACL.
- Min, S., et al. (2023). *FActScore: Fine-grained Atomic Evaluation of Factual Precision in Long Form Text Generation*. EMNLP.
- Zhang, T., et al. (2020). *BERTScore: Evaluating Text Generation with BERT*. ICLR.

13 Appendix A. Reproducibility

- Repo: Akshitha024/legal-summarization-benchmark, MIT.
- Reproduce: `make eval` & `make plots`.
- 5 figures in `results/figures/`.
- Test artifacts in `docs/test_results/`.